Fusion Equilibrium Challenge · Getting Started Kit

NeurIPS 2026 Competition Track · A Sophelio initiative

This kit gets you from zero to your first submission. No fusion background required. Follow the steps in order — you can make a first submission in under an hour.

Step 1 Understand the problem

Reconstruct the magnetic geometry of a fusion plasma using only non-magnetic diagnostics.

Your inputs	Your targets
· Poloidal-field coil currents · Plasma current (Ip) · Thomson-scattering Te and ne profiles · Fixed machine geometry	· Full 65×65 poloidal flux map ψ(R,Z) · Last closed flux surface (LCFS) · Five scalar parameters: βN, li, q95, Raxis, Zaxis

Step 2 Download dFL and explore the data

Before writing any training code, explore what you're working with. dFL is Sophelio's free desktop platform — it renders the ψ(R,Z) flux map, Thomson-scattering profiles, and coil-current traces for every shot so you can see exactly what your model is learning to predict.

> **Download dFL free — no account required**
>
> 🔗 dfl.sophelio.io · macOS · Windows · Linux
>
> *Tip: open a DIII-D shot, then a MAST shot. Notice how different the plasma geometry looks — that difference is the cross-machine challenge.*

Step 3 Access the dataset

The full dataset is on Hugging Face under CC BY 4.0 — 9,121 shots, 98 GB.

```
from datasets import load_dataset
ds = load_dataset("Sophelio/fusion-equilibrium-challenge")
```

🔗 **Dataset:** huggingface.co/datasets/Sophelio/fusion-equilibrium-challenge

Six sample shots ship with the starter kit — start there before downloading the full corpus.

Step 4 Run a reference baseline

Four baselines are provided — from ridge regression to a convolutional decoder. Run one before building your own model.

🔗 **Baselines & code:** github.com/fusion-equilibrium-challenge

Estimated time from setup to first leaderboard submission using a baseline: 1–2 hours.

Step 5 Build your submission

Each submission has two components:

→ Harmonization layer (H) — transforms raw multi-rate signals into a reproducible tensor
→ ML model (M) — predicts the flux map, LCFS, and scalar parameters

Submissions are cross-evaluated: each model M is tested against all other teams' harmonization layers H. Data preparation quality is scored independently from model quality.

🔗 **Leaderboard & submit:** fusion-equilibrium-challenge.sophelio.io

Step 6 Skills that win

Get you started	Win the awards
· Python	· Neural operators (FNO, DeepONet)
· PyTorch or TensorFlow	· Transfer learning / domain adaptation
· Time-series modeling	· Physics-informed ML
· Data preprocessing / harmonization	· Image-to-image regression / conv. architectures

TIMELINE & AWARDS

Phase 1 — open now	Now → October 18, 2026
Phase 2 — blind test	October 19–26, 2026
NeurIPS 2026 results	December 2026

→ $500 — Best DIII-D intra-machine reconstruction
→ $500 — Best zero-shot DIII-D → MAST cross-machine generalization
→ Named co-authorship on the post-competition lessons-learned paper
→ Invited talk at the NeurIPS 2026 Competition Workshop

ALL LINKS

🔗 **Download dFL:** dfl.sophelio.io
🔗 **Challenge website:** fusion-equilibrium-challenge.sophelio.io
🔗 **Dataset:** huggingface.co/datasets/Sophelio/fusion-equilibrium-challenge
🔗 **Baselines & code:** github.com/fusion-equilibrium-challenge
🔗 **Announcement:** sophelio.io/fusion-equilibrium-challenge-neurips-2026-competition
✉ **Questions:** challenge@sophelio.io